Lecture 23.14: Evaluation Metrics for Regression

📌 Topic Overview

In this concluding lecture for Day 23, Shraddha ma'am covers all essential **Regression Evaluation Metrics**: **MAE**, **MSE**, **RMSE**, R^2 **Score**, and **Adjusted** R^2.

🏛️ Comprehensive Metrics Formulations

1. Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{m}\sum_{i=1}^{m}|y_i - \hat{y}_i|$$

- In the original units of y. Robust to extreme outliers.

2. Mean Squared Error (MSE) & Root Mean Squared Error (RMSE)

$$\text{MSE} = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2, \quad \text{RMSE} = \sqrt{\text{MSE}}$$

- RMSE converts the squared metric back into the original units of the target variable.

3. R^2 Score (Coefficient of Determination)

$$R^2 = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

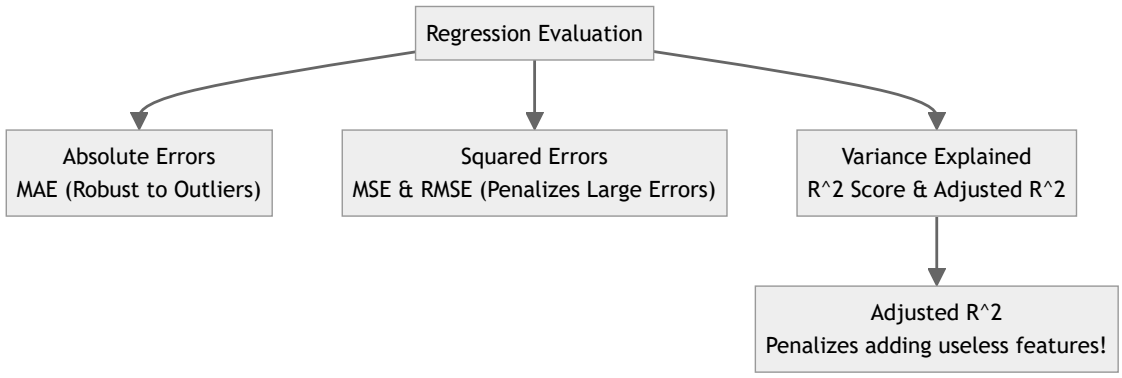
- Measures the proportion of variance in y explained by the features (1.0 is perfect; 0.0 equals baseline mean predictor).

4. Adjusted R^2

$$R_{\text{adj}}^2 = 1 - \left[\frac{(1 - R^2)(n - 1)}{n - p - 1}\right]$$

- **Why Adjusted R^2?** Adding irrelevant garbage features will *always* mechanically inflate raw R^2. Adjusted R^2 penalizes model complexity by including feature count p in the denominator!

📊 Metrics Flowchart



💻 Python Implementation

```
import numpy as np
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

def compute_regression_metrics(y_true, y_pred, num_features):
    """
    Computes all standard regression evaluation metrics:
    1. MAE (Mean Absolute Error)
    2. MSE (Mean Squared Error)
    3. RMSE (Root Mean Squared Error)
    4. R^2 (Coefficient of Determination)
    5. Adjusted R^2 (Penalizes redundant features)
    """
    y_t = np.array(y_true)
    y_p = np.array(y_pred)
    n = len(y_t)
    p = num_features

    mae = mean_absolute_error(y_t, y_p)
    mse = mean_squared_error(y_t, y_p)
    rmse = np.sqrt(mse)
```

```
    r2 = r2_score(y_t, y_p)

    # Adjusted R^2 formula
    adj_r2 = 1.0 - ((1.0 - r2) * (n - 1) / (n - p - 1))

    return {
        'MAE': float(mae),
        'MSE': float(mse),
        'RMSE': float(rmse),
        'R2_Score': float(r2),
        'Adjusted_R2': float(adj_r2)
    }

# --- Function Call & Testing ---
actuals = [100.0, 150.0, 200.0, 250.0, 300.0]
preds   = [105.0, 142.0, 208.0, 245.0, 310.0]

metrics = compute_regression_metrics(actuals, preds, num_features=2)
print("Regression Evaluation Metrics:")
print(f"  MAE         : {metrics['MAE']:.2f}")
print(f"  MSE         : {metrics['MSE']:.2f}")
print(f"  RMSE        : {metrics['RMSE']:.2f}")
print(f"  R^2 Score   : {metrics['R2_Score']:.4f}")
print(f"  Adjusted R^2: {metrics['Adjusted_R2']:.4f}")
```

🔑 Key Takeaways

1. MAE measures average magnitude; RMSE penalizes large blunders.
2. R^2 tells you what percentage of target variance your model captures.
3. Always report Adjusted R^2 when performing feature selection to detect useless predictors.